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PERCUTANEOUS TARGETING DEVICE

Abstract

A percutaneous targeting device comprises a body holding an extension. The body has a target guide holding a target pin which may be a K-wire. The extension has two guide sleeves, holding a guide wire. All wires are held in a common plane. The guide wires serve as a guide for cannulated fixation screws. The extension is movable against the body to adjust the angle between the target pin and the guide wires.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/890,350 filed Aug. 18, 2022, which is a continuation of U.S. patent application Ser. No. 17/885,803 filed Aug. 11, 2022, which is a continuation of U.S. patent application Ser. No. 17/252,865 filed Dec. 16, 2020, which is a United States National Phase application of PCT Application No. PCT/US2019/038116 filed Jun. 20, 2019, which claims priority to European Patent Application Ser. No. 18/178,849.8 filed on Jun. 20, 2018.

BACKGROUND

[0002] This disclosure relates to a percutaneous targeting device which may be used in a chevron osteotomy procedure.

[0003] A hallux valgus deformity is a common foot disorder that can result in the formation of a bunion on a patient's foot. A known procedure for the correction of a hallux valgus deformity is the minimal-invasive chevron osteotomy procedure. After performing the chevron osteotomy, bone fragments are realigned and fixed by fixation screws. The screws may be inserted by the surgeon in a minimally invasive fashion in an oblique angle into the bone. For correct placement of the screws, a large number of X-ray images is required which may result in loss of time, a large radiation dose, and costs.

SUMMARY

[0004] Disclosed herein are guides to aid in screw placement for minimally invasive foot surgery (e.g., bunion surgery-Chevron and/or Akin osteotomies). Devices herein ease guide wire (e.g., Kirschner-wire, also known as a K-wire) placement for screw fixation (e.g., cannulated screw)

[0005] The percutaneous targeting device, herein also mentioned as tool, according to the embodiments allows simple and precise targeting for the placement of screws to secure bone fragments of small bones. After a target pin has been placed in a bone (which may be made by a drill or similar tool) and the tool is placed on the reference pin, the tool ensures that the screws are in a plane through the target pin. Due to markings and/or a window, the length of the target pin can easily be adjusted such that the target pin ends within a target window or at an intersect point.

[0006] Location of a plane can easily be checked by a surgeon by considering the position of the body and the target pin. For better indicating the plane, the body may have a flat shape. The tool can be adjusted by moving the extension relative to the body, such that the angle between the target pin and the guide pins can be adapted to specific needs. Independent of this adjustment, the guide pins remain parallel. If one of the guide pins is fixed to the bone (which may be made by a drill or similar tool), the tool is fixed in its position relative to the bone. The second or any further guide pins are forced to be parallel to the first guide pin. Therefore, oblique screws and the weakening of bone material are prevented. The length of the guide pins is indicated by markings at the tool. The length of the first guide pin to contact with its tip the tip of the target pin is always the same if the

target pin has been adjusted with respect to the tool or vice-versa as mentioned above. The indicated lengths allow a calculation of the required screw lengths.

[0007] A method of fixing bone fragments in minimally invasive foot surgery comprises using a percutaneous targeting device. The targeting device with the target guide can be positioned on the target pin, such that the target pin is guided within the target guide. Positioning may be done such that a visible mark at the target pin is aligned with the body of the target guide or a marking thereof. Alternatively, the sleeve of the target guide may be positioned over the target pin and the targeting device body may be attached later to the target guide. After performing the osteotomy, the hook of a hook sleeve assembly may be inserted intramedullary into the proximal part of the metatarsal bone. A first guide wire may be placed through a positioning screw of the hook sleeve assembly and the hook sleeve into the distal part of the metatarsal bone. The distal part of the metatarsal bone may be shifted/positioned by rotating the positioning screw along the guide wire. The tip of the positioning screw may be in direct contact with the distal bone fragment or may push the fragment through the skin (via a plunger tip). Additionally, a first guide sleeve may be inserted into the outmost attachment hole and at least a further guide sleeve may be inserted into one of the other attachment holes. Then, the tool can be adjusted by moving the extension such that the guide sleeves or guide wires are under a desired angle relative to the target pin. This allows a precise selection of the screw angle in the bone.

[0008] By these methods, multiple screws are precisely parallel. The length of the screws can be selected based on the lengths of the first guide wire and the second guide wire extending into the bone. Therefore, perforating two surfaces of the bone by screws that are too long can be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows a targeting device.

[0010] FIG. 2 shows the application of the targeting device at a bone.

[0011] FIG. 3 shows details of the geometry and movement of the targeting device.

[0012] FIGS. 4a, 4b, and 4c show a further embodiment of the targeting device.

[0013] FIG. 5 shows a sectional view of the targeting device.

[0014] FIG. 6 shows a perspective view of the targeting device.

[0015] FIG. 7 shows a perspective view of a further targeting device.

[0016] FIG. 8 shows a hook sleeve.

[0017] FIG. 9 shows a detailed view of a hook sleeve.

DETAILED DESCRIPTION

[0018] Embodiments of devices disclosed herein simplify the placement of fixation screws in a bone of distal extremities, which can be problematic during such surgical procedures. For example, fixation of metatarsal bone fragments is simplified using the devices as described herein after a minimally invasive chevron osteotomy procedure.

[0019] In a first embodiment, a percutaneous targeting device for minimal invasive foot surgery comprises a body holding an extension. The body has attachment means for a target guide comprising a target sleeve for holding a target pin. Additionally or alternatively, a hook sleeve or hook sleeve assembly having a sideward hook may be used. The extension has multiple attachment holes for holding at least one guide wire and/or at least one guide sleeve. The at least one guide sleeve may further hold a guide wire. The guide sleeves give a better guidance and a higher stability to the guide wires. The target sleeve is configured for holding a target pin, which may be a K-wire. The target pin may extend axially through the target sleeve. The target sleeve may be configured to hold the target pin in a fixed position relative to the target sleeve. If required, the target pin may be released from its fixed position. The target pin may be movable within the target

sleeve, in a direction, which may be along its longitudinal axis. This may allow to position the target pin and/or to insert/remove the target pin. The guide sleeves may be configured for holding guide wires. A K-wire is a wire having a tip and configured to be drilled into a bone. The guide wires may have a tip and may be configured to be drilled into a bone. Any of the guide wires may be a K-wire. The target pin and the guide wires may have a diameter between 5 mm and 0.5 mm or between 3 mm and 1 mm. A first attachment hole for holding a first guide sleeve or a first guide wire is located out most of the extension and therefore most distant from the target sleeve. Further attachment holes for holding further guide sleeves or guide

[0020] Wires may be parallel to the first attachment hole and closer to the target sleeve. The extension may be movable against the body and may be held slidably in the body to extend the distance and/or angle between the target sleeve and the guide sleeves. The target sleeve and the guide sleeves and/or the guide wires may be in a common plane and point into a common target area. Therefore, wires held by the target sleeve and the guide sleeves may be in the same plane and when inserted correctly, their tips may end within the target area.

[0021] The target sleeve may have a tubular shape forming a target sleeve channel. The first guide sleeve may have a tubular shape forming a first guide sleeve channel. A second guide sleeve may have a tubular shape forming a second guide sleeve channel. The other guide sleeves may have a tubular shape, and the guide sleeves forming a guide sleeve channel. The channels may have an inner bore adapted to the outer diameter of the wires they are used for. The target guide may be adapted to fit to the body and the guide sleeves may be adapted to fit into the attachment holes. The guide sleeves may have marks or rules which may indicate the depth of wires held by the guide sleeves. For this purpose, the wires may have visible marks such as laser marks.

[0022] The body and the extension may be arc-shaped. The arc-shaped body may define a first radial axis and the arc-shaped extension may define a second radial axis. The first radial axis and the second radial axis may intersect at an intersect point within the target area. The target sleeve and one of the guide sleeves may also point to an intersect point within the target area which may be the same intersect point as mentioned above. The target area may be defined as a circular area centered on the intersect point. When sliding the extension in or out of the body, there may be an arched movement and the intersect point may not move, but the angle between the target sleeve channel and the first guide sleeve channel or a first guide wire, if the guide wire is held without sleeve, changes.

[0023] The target pin may have at least one visible mark which may be a laser mark in a predetermined distance from its tip. The target guide may have a window through which the target pin may be seen such that the visible mark at the target pin may be aligned with the body. For this purpose, the body may have an alignment mark. Alignment of the visible mark at the target pin with the body may ensure that the tip of the target pin is within the target area. In a further embodiment, the tip of the target pin may be at the intersect point. The target guide may also have a target sleeve which may have a coupling section, which may be threaded and by which it may be moved to position the metatarsal bone fragments. The target sleeve may have a gear rack which interfaces with a rack-wheel at the body and which allows movement of the target sleeve along a longitudinal axis of the target sleeve.

[0024] In a further embodiment, the body may be one piece with the extension or at least fixedly attached thereto. Angular adjustment may be made by selecting attachment holes which result in a required angle between the target pin and the at least one guide wire.

[0025] The percutaneous targeting device, herein also mentioned as tool, according to the embodiments allow simple and precise targeting for the placement of screws to secure bone fragments of small bones. After the target pin has been placed in a bone (which may be made by a drill or similar tool) and the tool is placed on the reference pin, the tool ensures that the screws are in a plane through the target pin. Due to markings and/or a window, the length of the target pin can easily be adjusted such that the target pin ends within a target window, or even at an intersect point.

[0026] The location of the plane can easily be checked by a surgeon by considering the position of the body and the target pin. For better indicating the plane, the body may have a flat shape. The tool can be adjusted by moving the extension relative to the body, such that the angle between the target pin and the guide pins can be adapted to specific needs. Independent of this adjustment, the guide pins remain parallel. If one of the guide pins is fixed to the bone (which may be made by a drill or similar tool), the tool is fixed in its position relative to the bone. The second or any further guide pins may be forced to be parallel to the first guide pin. Therefore, oblique screws and the weakening of bone material are prevented. The length of the guide pins is indicated by markings at the tool. The length of the first guide pin to contact with its tip the tip of the target pin is always the same if the target pin has been adjusted with respect to the tool or vice-versa as mentioned above. The indicated lengths allow a calculation of the required screw lengths.

[0027] Finally, the screws, (e.g., cannulated fixation screws) may be inserted over the guide wires. The instrument can easily be removed by removing one wire after the other.

[0028] A method of fixing bone fragments in minimally invasive foot surgery comprises placing a target pin, which may be a K-wire, in the distal part of a metatarsal bone. Then an osteotomy may be performed, and the distal bone part with the target pin may be repositioned. In a next step, the targeting device with the target guide may be positioned on the target pin, such that the target pin is guided within the target guide. Positioning may be done such that a visible mark at the target pin is aligned with the body of the target guide or a marking thereof. Alternatively, the sleeve of the target guide may be positioned over the target pin, and the targeting device body may be attached later to the target guide. In a further embodiment, after performing the osteotomy, the hook of a hook sleeve assembly may be inserted intramedullary into the proximal part of the metatarsal bone. A first target pin may be placed through a positioning screw of the hook sleeve assembly and the hook sleeve into the distal part of the metatarsal bone. The distal part of the metatarsal bone may be shifted/positioned by rotating the positioning screw along the target pin. The tip of the positioning screw may be in direct contact with the distal bone fragment or may push the fragment through the skin (e.g., via a plunger tip).

[0029] A first guide sleeve may be inserted into the outmost attachment hole and at least a further guide sleeve may be inserted into one of the other attachment holes. This step may also be performed before positioning the tool over the reference guide. The guide sleeves may also be pre-assembled. The tool can be adjusted by moving the extension such that the guide sleeves or guide wires may be under a desired angle relative to the target pin. This allows a precise selection of the screw angle in the bone.

[0030] Guide wires, e.g., two guide wires, may be inserted through the guide sleeves into the bone. Alternatively, at least one guide wire may be directly inserted through an attachment hole. In this embodiment, the step of inserting a guide sleeve can be omitted. The tip of the outmost guide wire in a first guide sleeve may contact the tip of the target pin within the target area or may even contact at the intersect point. The tip of other guide wires in other guide sleeves may contact a target pin within the target area. An intersection between the target pin and the first guide wire and/or an intersection between the target pin and the second guide wire may be outside of the target sleeve. The target pin and the guide wires may be all in the same plane. The guide wires may be inserted in any sequence. The first guide wire may be inserted first, followed by the second guidewire. Alternatively, the first guide wire may follow the second guide wire. It may also be possible to insert a guidewire first, followed by a target pin.

[0031] Any of the target pin, the first guide wire, and the second guide wire may define a center axis. In general, the center axis of the target pin may intersect with the center axis of the first guide wire and/or the center axis of the target pin may intersect with the center axis of the second guide wire. The center axis of the first guide wire may be parallel to the center axis of the second guide wire.

[0032] In an embodiment, the extension may be adjusted to bring the guide sleeve in a new

position for another guide wire after inserting a first guide wire.

[0033] Cannulated fixation screws may be screwed into the bone by using the guide wires. The guide wires pass through the cannulations of the screws. The tool may be removed before or after screwing the fixation screws into the bone. At the end, the tool, the guide wires, and the target pin may be removed.

[0034] By these methods, multiple screws are precisely parallel. The length of the screws can be selected based on the lengths of the first guide wire and the second guide wire extending into the bone. Therefore, perforating two surfaces of the bone by screws that are too long can be prevented.

[0035] In FIG. 1, a first embodiment is shown. The percutaneous targeting device **100** comprises a body **110** and an extension **120**. The extension may be movable with respect to the body **110** for angular adjustment. The body **110** has attachment means for a target guide **200**. In an embodiment, the target guide **200** has a tubular target sleeve **210** with a target sleeve channel **211** having a diameter adapted to hold a target pin **220**. There may be a hole or a cylindrical bore in the device body **110** to hold the target guide **200**. There may also be locking means **112** at the device body **110** for fixing the target guide **200** at its position. The target guide **200** may be moveable within the body **110** and it may be removable from the body **110**. The target guide **200** may also be fixed to the body **110** or one part with the body **110**. It may further have a window **214** through which a mark **222** such as a laser mark at the target pin **220** can be seen aligned with the body **110** or a mark at the body **110** or a mark at the target guide **200**. The target guide **200** may also have a coupling section **212** such as a threaded section by which it may be held by a handle for easier installation. A target pin **220** may be a K-wire having a tip and configured to be drilled into a bone.

[0036] The extension **120** may have multiple attachment holes for holding guide sleeves **310**, **330**. A guide sleeve **310**, **330** may have a tubular body forming a channel **311**, **331** for holding a guide wire. There may be a first attachment hole **121** at an outmost position, most distant from the target guide **200**. There may be an attachment opening **125** including further attachment holes **122**, **123**, **124** with increasing distances from the first attachment hole, but closer to the target guide **200**. The attachment holes **121**, **122**, **123**, **124** may include cylindrical bores, but any other suitable form may be used. The attachment holes **122**, **123**, **124** may be so close to one another that there exist no sidewalls between the holes. There may be at least one guide sleeve **310** in the outmost or first attachment hole **121**. There may be a second guide sleeve **330** in one of the other attachment holes **122**, **123**, **124**. The guide sleeves **310**, **330** may be moveable within the extension **120**, and they may be removable from the extension **120**.

[0037] An appropriate attachment hole **121**, **122**, **123**, **124** may be selected depending on the desired distance from between a first guide wire **320** held by the first guide sleeve **310** to a second guide wire **340** held by the second guide sleeve **330**. The guide wires **320**, **340** may have a tip and may be configured to be drilled into a bone. The first guide wire **320**, the second guide wire **340**, and the target pin **220** may be all in the same plane. When extended properly, the tip of the target pin **220** may contact the tip of the first guide wire **320**. The tip of the second guide wire **330** may contact the target pin **220** distant from its tip, but still in the same plane. Herein the term of the same plane may include deviations with a maximum of 5 mm, 2 mm, or 1 mm, for example.

[0038] The guide sleeves **310**, **330** may have markings or depth scales **312**, **332** to indicate the depth or length of an inserted guide wire **320**, **340**. For a precise length reading, the guide wires **320**, **340** may also have marks, for example laser marks.

[0039] In an embodiment, the target sleeve channel **211**, the first guide sleeve channel **311**, and the second guide sleeve channel **331** may be configured such that the intersection between the target pin **220** and the first guide wire **320** and/or the intersection between the target pin **220** and the second guide wire **340** is outside of the target sleeve **210**.

[0040] In FIG. 2, the application of the targeting device **100** at a bone is shown. The big toe **500** comprises a metatarsal bone **530**, a proximal phalanx **520**, and a distal phalanx **510**. The metatarsal bone **530** has been cut such that there exist two fragments **531**, **532**. These two fragments have to

be screwed together in a new position as shown by two bone screws. Here, the target pin **220** is inserted into the metatarsal bone fragment **531** and holds the guide tool. A first guide wire **320** and a second guide wire **340** have already been inserted. A first cannulated screw **410** and a second cannulated screw **420** have already been screwed into the bone to fix the fragments **531** and **532** together.

[0041] In FIG. 3, details of the geometry and movement of the targeting device are shown. The body **110** may be arc shaped and may have a hollow opening **115** to receive at least part of the extension **120**, which also may be arc shaped. The extension **120** may be configured to move within device body **110**. The extension **120** may be configured to move into a direction **172** relative to the body **110** for a rotational adjustment. This changes the rotation angle **176** between the center axis **161** of the target pin **220** and the center axis **162** of the first guide wire **320**. The target pin **220** may be in fixed angular relationship with the device body **110** and the guide wires **320**, **340** may be in fixed angular relationship with the device extension **120**. The center axis **161** of the target pin **220** and the center axis **162** of the first guide wire **320** intersect within a target area **155** or even at an intersect point **163** independent of the relative position of the extension **120** relative to the body **110**.

[0042] For size adjustment, the position of the target pin **220** may be adjusted in a direction **171** along its center axis **161**. Depth adjustment may be made by moving any of the guide wires **320**, **340** in a direction **174** along or parallel to the first guide wire **320** center axis **161**. Distance adjustment of the guide wires **320**, **340** may be made by a parallel movement of the guide wires into direction **173**.

[0043] The target pin **220** and the first guide wire **320** may contact within the target area **155** or even at the intersect point **163**. The target pin **220** and the second guide wire **340** may contact within the target area **155**. The target area may be circular and have a diameter of 10 mm, 5 mm, or 2 mm, for example.

[0044] In an embodiment, the extension **120** may be configured to move relative to the body **110** such that the point of intersection between the target pin **220** and the first guide wire **320** does not move.

[0045] In FIGS. 4a, 4b, and 4c, a further embodiment is shown. Here, the first guide wire **320** and the second guide wire **340** may be held by the extension **120** without sleeves, and a third guide wire **360** may be held by a guide sleeve **350**. The third guide wire **360** may be parallel to and in the same plane with the first guide wire **320** and the second guide wire **340**. The center axis of the guide sleeves and/or the center axis of the guide wires may be in the same plane.

[0046] In this embodiment, the target sleeve **210** may have a gear rack **216** at one side which interfaces with a rack-wheel **217** shown in FIG. 5. The rack-wheel **217** may be driven by a handle **116**. Rotation of the handle **116** may rotate the rack-wheel **217** and therefore move the gear rack **216** and the target sleeve **210** in and out.

[0047] There may be an extension lock **119** to lock the device extension **120** in a fixed relationship to the device body **110**. The extension lock **119** may be a button which releases the device extension **120** when pressed, such that the device extension **120** may be moved relative to the device body **110**.

[0048] The features of the embodiments shown herein may be combined in any combination.

[0049] Before inserting any guide wire into the bone, the targeting device may be rotated around the target pin **220** in a direction **175** which results in an angulation in an angulation angle **177**.

[0050] FIG. 4a shows a front view, FIG. 4b shows a right side view, and FIG. 4c shows a top view of the embodiment.

[0051] FIG. 5 shows a sectional view of the targeting device.

[0052] In this embodiment, any guide wire **320**, **340**, **360** may be directly held by the device extension **120** without using a guide sleeve **310**, **330**, **350**. For this purpose, a holding structure **126** may be in the device extension **120**. This holding structure **126** may include holes, ribs, wedges or

other elements forming attachment holes which may be adapted to hold guide wires and/or sleeves. [0053] FIG. 6 shows a perspective view of the targeting device. [0054] FIG. 7 shows a perspective view of a further targeting device. This embodiment has a partially simplified body and a hook sleeve assembly comprising a positioning screw **610** and a hook sleeve **615** further comprising a hook. Here, at least some of the sleeves may be integrated into the body and/or extension. [0055] FIG. 8 shows a hook sleeve and a positioning screw in detail. A positioning screw **610** having an inner sleeve **613** fits on the target pin **220**. The positioning screw **610** may have an outer thread **612** and may have a nut **611**. The positioning screw **610** may be screwed into a hook sleeve **615**. When screwed together, the sleeve assembly comprising the inner sleeve **613** and the positioning screw **610** may firmly hold at the target pin **220**, which may be close to the tip of the target pin **220**. The hook sleeve **615** may have a hook **616** with an extension radial to the target pin **220**. The hook **616** may further have an axial extension to provide an offset in axial direction. The hook **616** may include a wire which may have a diameter of less than 2 mm or less than 1 mm and may have a length of less than 15 mm or less than 10 mm. The hook sleeve **615** may be used as an alternative to the target sleeve **210**. [0056] FIG. 9 shows a detailed view of a hook sleeve assembly. Here, the hook sleeve **615** and the positioning screw **610** are disassembled.

LIST OF REFERENCE NUMERALS

[0057] **100** targeting device [0058] **110** device body [0059] **112** locking means [0060] **115** hollow opening [0061] **116** handle [0062] **119** extension lock [0063] **120** device extension [0064] **121** first attachment hole [0065] **122, 123, 124** attachment hole [0066] **126** holding structure [0067] **155** target area [0068] **161** target pin center axis [0069] **162** first guide wire center axis [0070] **163** intersect point [0071] **171** direction of size adjustment by the target pin [0072] **172** direction of rotational adjustment [0073] **173** direction of distance adjustment [0074] **174** direction of depth adjustment [0075] **175** direction of angulation adjustment [0076] **176** rotation angle [0077] **177** angulation angle [0078] **200** target guide [0079] **210** target sleeve [0080] **211** target sleeve channel [0081] **212** coupling section [0082] **214** window [0083] **216** gear rack [0084] **217** rack wheel [0085] **220** target pin [0086] **222** laser mark [0087] **310** first guide sleeve [0088] **311** first guide sleeve channel [0089] **312** depth scale [0090] **320** first guide wire [0091] **330** second guide sleeve [0092] **331** second channel [0093] **332** depth scale [0094] **340** second guide wire [0095] **350** third guide sleeve [0096] **360** third guide wire [0097] **410** first screw [0098] **420** second screw [0099] **500** big toe [0100] **510** distal phalanx [0101] **520** proximal phalanx [0102] **530** metatarsal bone [0103] **531,532** metatarsal bone fragments [0104] **610** hook sleeve assembly [0105] **611** nut [0106] **612** thread [0107] **613** inner sleeve [0108] **615** hook sleeve [0109] **616** hook

Claims

1. A percutaneous targeting device comprising: a body; an extension moveable relative to the body, the extension including an attachment opening having a first attachment hole and a second attachment hole, and no sidewall is located between the first attachment hole and the second attachment hole, and the extension includes another attachment hole; a targeting assembly including a target channel; a target pin receivable through the body and configured to be placed in a distal bone part of a bone to reposition the distal bone part of the bone, wherein the target pin is received through the target channel of the targeting assembly; a guide wire insertable through one of the first attachment hole and the second attachment hole and configured to be inserted into the bone; another guide wire inserted through the another attachment hole and configured to be inserted into the bone; and a cannulated fixation screw configured to be screwed into the bone using the guide wire.
2. The targeting device as recited in claim 1, wherein the bone is a metatarsal bone that is split into

the distal bone part and a proximal bone part.

3. The targeting device as recited in claim 1, wherein the guide wire is received in one of the first attachment hole and the second attachment hole depending on a desired distance between the guide wire and the another guide wire.
4. The targeting device as recited in claim 1, including another cannulated fixation screw configured to be screwed into the bone using the another guide wire.
5. The targeting device as recited in claim 1, wherein the attachment opening is between the targeting assembly and the another attachment hole.
6. The targeting device as recited in claim 1, wherein the another attachment hole is most distal to the target pin and the attachment opening is most proximal to the target pin.
7. The targeting device as recited in claim 1, wherein the targeting assembly includes a target sleeve that defines the target channel that receives the target pin.
8. The targeting device as recited in claim 7, wherein the target sleeve is configured to allow for movement of the target pin within the targeting assembly.
9. The targeting device as recited in claim 1, wherein a position of the extension is adjustable relative to the body to change an angle between the target pin and the guide wire.
10. The targeting device as recited in claim 1, wherein the cannulated fixation screw is a non-compressive cannulated fixation screw.
11. The targeting device as recited in claim 1, wherein the targeting assembly is configured to hold the target pin in a fixed position relative to the targeting assembly.
12. The targeting device as recited in claim 1, wherein the target pin is a K-wire that extends axially through the targeting assembly.
13. The targeting device as recited in claim 1, wherein the guide wire and the another guide wire each have a tip and is configured to be drilled into the bone.
14. The targeting device as recited in claim 1, wherein the body is arc-shaped.
15. The targeting device as recited in claim 1, wherein the attachment opening includes a third attachment hole, and no sidewalls are located between the first attachment hole, the second attachment hole, and the third attachment hole.
16. The targeting device as recited in claim 1, wherein a periphery of the targeting assembly includes a window along the target channel, the target pin includes a first indicator, the body includes a second indicator, and the first and second indicators are alignable with each other at a position along the window.
17. A targeting device comprising: a body; an extension moveable along an arc path established by the body, the extension including a first attachment hole and an attachment opening having a row of overlapping attachment holes; a targeting assembly coupled to the body, wherein the targeting assembly includes a target channel dimensioned to receive a target pin that is insertable in a first bone part to reposition the first bone part relative to a second bone part; a first guide sleeve insertable in the first attachment hole, the first guide sleeve dimensioned to receive a first guide wire insertable in the second bone part; and a second guide sleeve insertable in a selectable one of the overlapping attachment holes, the second guide sleeve dimensioned to receive a second guide wire insertable in the second bone part; wherein the first and second guide wires establish a trajectory of respective cannulated fixation screws adapted to secure the first bone part and the second bone part to each other.
18. The targeting device as recited in claim 17, wherein the targeting assembly includes a target sleeve captured in a passageway of the body, the target sleeve establishes the target channel, and the target sleeve is translatable along an axis of the passageway.
19. The targeting device as recited in claim 18, wherein a distal end of the target sleeve is adapted to engage a periphery of the first bone part.
20. The targeting device as recited in claim 18, further comprising: a drive mechanism, wherein actuation of the drive mechanism causes the target sleeve to translate along the axis.

